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### Inductor component and mounting component

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#### Abstract

An inductor component includes an element body having a main surface, a coil disposed in the element body, and extending along the main surface, and a first vertical wiring and a second vertical wiring provided in the element body such that respective end surfaces thereof are exposed from the main surface of the element body, and connected to respective ends of the coil. The coil has a coil wiring portion having a first end portion and a second end portion, and extending along a circumferential direction, with the number of turns equal to or less than 1, a first pad portion that is a portion to which the first vertical wiring is connected and that is connected to the first end portion, and a second pad portion that is a portion to which the second vertical wiring is connected and that is connected to the second end portion.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims benefit of priority to Japanese Patent Application No. 2021-180474, filed Nov. 4, 2021, the entire content of which is incorporated herein by reference.

### BACKGROUND

#### Technical Field

(2) The present disclosure relates to an inductor component and a mounting component.

#### Background Art

(3) There has been an inductor component described in Japanese Unexamined Patent Application Publication No. 2014-13815. The inductor component includes a substrate, spiral conductors provided on both surfaces of the substrate, respectively, a magnetic layer covering the spiral conductor, an outer electrode provided on a surface of the magnetic layer, an extended conductor connected to an outer peripheral end of the spiral conductor, and a connection conductor connecting the extended conductor and the outer electrode. The spiral conductor and the extended conductor constitute a coil. The extended conductor is disposed outside in a radial direction of the spiral conductor, and extends along a side of the substrate.

### SUMMARY

(4) Incidentally, in the above-described existing inductor component, when a DC superposition allowable current is increased in order to cause a large current to flow, the number of turns of the

spiral conductor is made to be equal to or less than 1 in order to suppress magnetic saturation. It has been found that there are the following problems when actually trying to manufacture such an inductor component.

(5) Since both ends of the spiral conductor come close to each other, the two extended conductors respectively connected to both the ends of the spiral conductor also come close to each other, and there is a possibility that the two extended conductors having a greatest potential difference in the coil will be short-circuited. In addition, since the two extended conductors are disposed outside in the radial direction of the spiral conductor, there is a possibility that the inductor component will become large.

(6) Accordingly, the present disclosure provides an inductor component and a mounting component capable of reducing a possibility of a short circuit of a coil and of reducing the size of the inductor component.

(7) An inductor component as an aspect of the present disclosure includes an element body having a main surface, a coil disposed in the element body, and extending along the main surface, and a first vertical wiring and a second vertical wiring provided in the element body such that respective end surfaces thereof are exposed from the main surface of the element body, and connected to respective ends of the coil. The coil has a coil wiring portion having a first end portion and a second end portion, and extending along a circumferential direction, with the number of turns equal to or less than 1, a first pad portion that is a portion to which the first vertical wiring is connected and that is connected to the first end portion, and a second pad portion that is a portion to which the second vertical wiring is connected and that is connected to the second end portion. At least a part of the first pad portion extends in a direction that intersects a direction in which the first end portion and the second end portion are connected and that is a direction in which an inner peripheral surface of the coil wiring portion is approached when viewed from a direction orthogonal to the main surface. At least a part of the second pad portion extends in the direction that intersects the direction in which the first end portion and the second end portion are connected and that is the direction in which the inner peripheral surface of the coil wiring portion is approached when viewed from the direction orthogonal to the main surface.

(8) Here, the fact that the number of turns of the coil wiring portion is less than 1 refers to a state in which the coil wiring portion does not have parts that are adjacent to each other in a radial direction of the coil and run side by side in a winding direction (that is, parts that are adjacent and overlap in the radial direction) when viewed from the direction orthogonal to the main surface. A fact that the number of turns of the coil wiring portion is equal to or greater than 1 refers to a state in which the coil wiring portion has parts that are adjacent to each other in the radial direction of the coil and run side by side in the winding direction when viewed from the direction orthogonal to the main surface. That is, a fact that the number of turns of the coil wiring portion is equal to or less than 1 refers to a state in which the above-described parts running side by side are not present, or both end surfaces of the coil wiring portion are tangent to the same plane along the radial direction of the coil when viewed from an axial direction of the coil.

(9) Further, a direction in which the first end portion and the second end portion are connected when viewed from the direction orthogonal to the main surface is a direction in which a center in a width direction of an end edge of the first end portion and a center in a width direction of an end edge of the second end portion are connected when viewed from the direction orthogonal to the main surface.

(10) According to the above-described aspect, the first pad portion extends in the direction that intersects the direction in which the first end portion and the second end portion are connected and that is a direction in which the inner peripheral surface of the coil wiring portion is approached, and the second pad portion extends in the direction that intersects the direction in which the first end portion and the second end portion are connected and that is a direction in which the inner peripheral surface of the coil wiring portion is approached. According to this, it is possible to

prevent the first pad portion and the second pad portion from approaching each other, and to reduce a possibility of a short circuit between the first pad portion and the second pad portion having a greatest potential difference in the coil. In addition, it is possible to prevent the first pad portion and the second pad portion from extending outward in the radial direction of the coil, to reduce the inductor component in size.

(11) Preferably, in an embodiment of the inductor component, the element body includes a magnetic powder and a resin containing the magnetic powder.

(12) According to the above-described embodiment, inductance can be increased.

(13) Preferably, an embodiment of the inductor component further includes a first external terminal in contact with the main surface and the end surface of the first vertical wiring, and a second external terminal in contact with the main surface and the end surface of the second vertical wiring.

(14) According to the above-described embodiment, since the first external terminal spreads across the main surface and the first vertical wiring, and the second external terminal spreads across the main surface and the second vertical wiring, sizes of the respective first and second external terminals can be controlled, and design of the first and second external terminals can be selected from a viewpoint of mounting properties and mechanical strength.

(15) Preferably, in an embodiment of the inductor component, the element body includes a magnetic powder, and the first external terminal and the second external terminal are in contact with the magnetic powder.

(16) According to the above-described embodiment, adhesion strength between each of the first external terminal and the second external terminal, and the element body is improved.

(17) Preferably, an embodiment of the inductor component further includes a coating film provided on the main surface, and the coating film includes a black colorant.

(18) According to the above-described embodiment, by providing the coating film on the main surface, when a plurality of the external terminals is provided on the main surface, insulation between the external terminals can be reliably ensured. In addition, since the coating film is black, it is possible to hide scratches on a surface of the element body, thereby improving a rate of good appearance.

(19) Preferably, in an embodiment of the inductor component, when viewed from the direction orthogonal to the main surface, the first pad portion has a linear first plane that is a distal end surface in an extending direction of the first pad portion and that faces the inner peripheral surface of the coil wiring portion, and a shortest distance between the first plane and the inner peripheral surface of the coil wiring portion is not a minimum among distances between an outer surface of the first pad portion and the inner peripheral surface of the coil wiring portion.

(20) Here, the distance between the outer surface of the first pad portion and the inner peripheral surface of the coil wiring portion refers to a shortest distance among distances between respective surfaces constituting the outer surface of the first pad portion and the inner peripheral surface of the coil wiring portion.

(21) According to the above-described embodiment, it is possible to reduce a possibility of a short circuit between the first plane of the first pad portion and the inner peripheral surface of the coil wiring portion.

(22) Preferably, in an embodiment of the inductor component, when viewed from the direction orthogonal to the main surface, the first pad portion has a linear first plane extending in the direction in which the first end portion and the second end portion are connected, and facing the inner peripheral surface of the coil wiring portion, and a linear second plane extending in a direction orthogonal to the direction in which the first end portion and the second end portion are connected, and facing the inner peripheral surface of the coil wiring portion. When a surface parallel to the first plane and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing the first plane is defined as a first virtual surface, and a surface parallel to the second plane and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing

the second plane is defined as a second virtual surface, a first distance  $a$  between the second virtual surface and the second plane is less than a second distance  $b$  between the first virtual surface and the first plane.

(23) According to the above-described embodiment, since the first distance  $a$  is less than the second distance  $b$ , it is possible to reduce a possibility of a short circuit between the first plane of the first pad portion and a part, of the inner peripheral surface of the coil wiring portion, facing the first plane. Even if a short circuit occurs between the second virtual surface and the second plane, a variation in a coil length is suppressed as compared to a case where a short circuit occurs between the first virtual surface and the first plane, and thus influence of a layer short can be suppressed.

(24) Preferably, in an embodiment of the inductor component, when viewed from the direction orthogonal to the main surface, the first pad portion has a linear first plane extending in the direction in which the first end portion and the second end portion are connected, and facing the inner peripheral surface of the coil wiring portion, and a linear second plane extending in a direction orthogonal to the direction in which the first end portion and the second end portion are connected, and facing the inner peripheral surface of the coil wiring portion. When a surface parallel to the first plane and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing the first plane is defined as a first virtual surface, and a surface parallel to the second plane and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing the second plane is defined as a second virtual surface, a first distance  $a$  between the second virtual surface and the second plane is equal to or greater than a second distance  $b$  between the first virtual surface and the first plane.

(25) According to the above-described embodiment, since the first distance  $a$  is equal to or greater than the second distance  $b$ , the first pad portion can be increased in the extending direction thereof. Thus, a size of the first pad portion can be increased, and connection reliability between the first pad portion and the first vertical wiring can be improved.

(26) Preferably, in an embodiment of the inductor component, the element body includes a magnetic powder, and the first distance  $a$  between the second virtual surface and the second plane is equal to or greater than twice a particle size D50 of the magnetic powder.

(27) According to the above-described embodiment, by increasing the distance between the second virtual surface and the second plane to be greater than the magnetic powder, it is possible to further reduce a possibility of a short circuit between the second virtual surface and the second plane.

(28) Preferably, in an embodiment of the inductor component, when viewed from the direction orthogonal to the main surface, a shortest distance  $c$  between the first pad portion and the second pad portion is greater than the first distance  $a$  between the second virtual surface and the second plane.

(29) According to the above-described embodiment, it is possible to further reduce a possibility of a short circuit between the first pad portion and the second pad portion that have a greatest potential difference in the coil. Even if a short circuit occurs between the second virtual surface and the second plane, a variation in the coil length is suppressed as compared to a case where a short circuit occurs between the first pad portion and the second pad portion, and thus influence of a layer short can be suppressed.

(30) Preferably, in an embodiment of the inductor component, when viewed from the direction orthogonal to the main surface, a shortest distance  $c$  between the first pad portion and the second pad portion is equal to or less than the first distance  $a$  between the second virtual surface and the second plane.

(31) According to the above-described embodiment, since the first pad portion and the second pad portion can be brought close to each other, a length of the coil wiring portion can be increased. Thus, the coil length can be increased, and the inductance can be increased.

(32) Preferably, in an embodiment of the inductor component, the element body includes a magnetic powder, and the shortest distance  $c$  between the first pad portion and the second pad

portion is equal to or greater than twice a particle size D50 of the magnetic powder.

(33) According to the above-described embodiment, by increasing the distance between the first pad portion and the second pad portion to be greater than the magnetic powder, it is possible to further reduce a possibility of a short circuit between the first pad portion and the second pad portion.

(34) Preferably, in an embodiment of the inductor component, the first pad portion has a first portion extending from the first end portion in the direction in which the first end portion and the second end portion are connected when viewed from the direction orthogonal to the main surface, and a second portion extending from a side of the first portion in the direction that intersects the direction in which the first end portion and the second end portion are connected and that is the direction in which the inner peripheral surface of the coil wiring portion is approached. The second pad portion has a first portion extending from the second end portion in the direction in which the first end portion and the second end portion are connected when viewed from the direction orthogonal to the main surface, and a second portion extending from a side of the first portion in the direction that intersects the direction in which the first end portion and the second end portion are connected and that is the direction in which the inner peripheral surface of the coil wiring portion is approached.

(35) According to the above-described embodiment, an area of the first pad portion can be increased, and the connection reliability between the first pad portion and the first vertical wiring can be improved. An area of the second pad portion can be increased, and connection reliability between the second pad portion and the second vertical wiring can be improved.

(36) Preferably, in an embodiment of the inductor component, when viewed from the direction orthogonal to the main surface, the first portion of the first pad portion does not protrude inward from the inner peripheral surface of the coil wiring portion.

(37) According to the above-described embodiment, it is possible to increase a distance between the inner peripheral surface of the coil wiring portion and the first pad portion in the direction in which the first end portion and the second end portion are connected, and to reduce a possibility of a short circuit between the coil wiring portion and the first pad portion in the direction in which the first end portion and the second end portion are connected.

(38) Preferably, in an embodiment of the inductor component, a thickness of the coil is less than a thickness of each of the first vertical wiring and the second vertical wiring.

(39) According to the above-described embodiment, volume of the element body can be increased, and the inductance can be increased.

(40) Preferably, an embodiment of the inductor component further includes a nonmagnetic insulating layer configured to cover at least a part of the coil.

(41) According to the above-described embodiment, insulation properties of the coil can be improved.

(42) Preferably, in an embodiment of the inductor component, the element body has a first magnetic layer and a second magnetic layer stacked in order in a first direction orthogonal to the main surface. The first vertical wiring and the second vertical wiring extend in the first direction from the respective ends of the coil. The first magnetic layer is present in a direction opposite to the first direction of the coil. The second magnetic layer is present in the first direction of the coil and in a direction orthogonal to the first direction, and the first vertical wiring and the second vertical wiring penetrate the second magnetic layer. A thickness of the first magnetic layer is greater than a thickness of the second magnetic layer.

(43) According to the above-described embodiment, since there is a limit in a thickness related to a process such as a resist thickness for each of the first vertical wiring and the second vertical wiring, there is a limit in the thickness of the second magnetic layer. On the other hand, the inductance can be increased by increasing the thickness of the first magnetic layer.

(44) Preferably, an embodiment of a mounting component includes a substrate, and the inductor

component according to the embodiment disposed in the substrate.

(45) According to the above-described embodiment, since the inductor component is built into the substrate, it is possible to increase a mounting area of a surface of the substrate.

(46) According to the inductor component and the mounting component as an aspect of the present disclosure, it is possible to reduce a possibility of a short circuit of a coil, and to reduce the size of the inductor component.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a plan view illustrating a first embodiment of an inductor component;
- (2) FIG. 2 is a sectional view taken along a line A-A in FIG. 1;
- (3) FIG. 3 is a plan view of the inductor component;
- (4) FIG. 4A is an explanatory view for explaining a method of manufacturing the inductor component;
- (5) FIG. 4B is an explanatory view for explaining the method of manufacturing the inductor component;
- (6) FIG. 4C is an explanatory view for explaining the method of manufacturing the inductor component;
- (7) FIG. 4D is an explanatory view for explaining the method of manufacturing the inductor component;
- (8) FIG. 4E is an explanatory view for explaining the method of manufacturing the inductor component;
- (9) FIG. 4F is an explanatory view for explaining the method of manufacturing the inductor component;
- (10) FIG. 4G is an explanatory view for explaining the method of manufacturing the inductor component;
- (11) FIG. 4H is an explanatory view for explaining the method of manufacturing the inductor component;
- (12) FIG. 4I is an explanatory view for explaining the method of manufacturing the inductor component;
- (13) FIG. 4J is an explanatory view for explaining the method of manufacturing the inductor component;
- (14) FIG. 4K is an explanatory view for explaining the method of manufacturing the inductor component;
- (15) FIG. 4L is an explanatory view for explaining the method of manufacturing the inductor component;
- (16) FIG. 4M is an explanatory view for explaining the method of manufacturing the inductor component;
- (17) FIG. 5A is a plan view illustrating an inductor component of a first comparative example;
- (18) FIG. 5B is a plan view illustrating an inductor component of a second comparative example;
- (19) FIG. 5C is a plan view illustrating an inductor component of a first working example;
- (20) FIG. 5D is a plan view illustrating an inductor component of a second working example;
- (21) FIG. 6 is a graph showing a relationship between relative values of an L value and extension amounts of pad portions in the first comparative example, the second comparative example, the first working example, and the second working example;
- (22) FIG. 7 is a graph of an enlarged part of FIG. 6;
- (23) FIG. 8 is a plan view illustrating a second embodiment of the inductor component;
- (24) FIG. 9 is a plan view illustrating a third embodiment of the inductor component; and

(25) FIG. 10 is a sectional view illustrating an embodiment of a mounting component.

## DETAILED DESCRIPTION

(26) Hereinafter, an inductor component and a mounting component as an aspect of the present disclosure will be described in detail by using embodiments illustrated in the figures. Note that, some of the figures are schematic, and do not reflect actual dimensions and ratios in some cases.

### First Embodiment

#### Configuration

(27) FIG. 1 is a plan view illustrating a first embodiment of an inductor component. FIG. 2 is a sectional view taken along the line A-A in FIG. 1.

(28) An inductor component **1** is mounted, for example, in electronic devices such as personal computers, DVD players, digital cameras, TVs, mobile phones, and car electronics, and is, for example, a component having a rectangular parallelepiped shape as a whole. However, the shape of the inductor component **1** is not particularly limited, and may be a cylindrical shape, a polygonal columnar shape, a truncated cone shape, or a truncated polygonal cone shape.

(29) As illustrated in FIG. 1 and FIG. 2, the inductor component **1** includes an element body **10**, a coil **15** disposed in the element body **10**, a nonmagnetic insulating layer **60** covering at least a part of the coil **15**, a first columnar wiring **31** and a second columnar wiring **32** provided in the element body **10** with respective end surfaces exposed from a first main surface **10a** of the element body **10** and connected to respective ends of the coil **15**, a first external terminal **41** exposed from the first main surface **10a** of the element body **10** and connected to the first columnar wiring **31**, a second external terminal **42** exposed from the first main surface **10a** of the element body **10** and connected to the second columnar wiring **32**, and a coating film **50** provided on each of the first main surface **10a** and a second main surface **10b** of the element body **10**.

(30) In the figure, a thickness direction of the inductor component **1** is defined as a Z direction, a forward Z direction is referred to as an upper side, and a reverse Z direction is referred to as a lower side. In a plane orthogonal to the Z direction of the inductor component **1**, a length direction that is a longer direction of the inductor component **1** and is a direction in which the first external terminal **41** and the second external terminal **42** are arranged is defined as an X direction, and a width direction of the inductor component **1** that is a direction orthogonal to the length direction is defined as a Y direction.

(31) The element body **10** has the first main surface **10a** and the second main surface **10b**, and a first side surface **10c**, a second side surface **10d**, a third side surface **10e**, and a fourth side surface **10f** that are located between the first main surface **10a** and the second main surface **10b** and that connect the first main surface **10a** and the second main surface **10b**.

(32) The first main surface **10a** and the second main surface **10b** are disposed on sides opposite to each other in the Z direction, respectively, the first main surface **10a** is disposed in the forward Z direction, and the second main surface **10b** is disposed in the reverse Z direction. The first side surface **10c** and the second side surface **10d** are disposed on sides opposite to each other in the X direction, respectively, the first side surface **10c** is disposed in a reverse X direction, and the second side surface **10d** is disposed in a forward X direction. The third side surface **10e** and the fourth side surface **10f** are disposed on sides opposite to each other in the Y direction, respectively, the third side surface **10e** is disposed in a reverse Y direction, and the fourth side surface **10f** is disposed in a forward Y direction.

(33) The element body **10** has a first magnetic layer **11** and a second magnetic layer **12** stacked in order along the forward Z direction. The term “in order” merely indicates a positional relationship between the first magnetic layer **11** and the second magnetic layer **12**, and has no relation to an order of formation of the first magnetic layer **11** and the second magnetic layer **12**. The forward Z direction is an example of a “first direction orthogonal to a main surface” described in the claims.

(34) Each of the first magnetic layer **11** and the second magnetic layer **12** includes a magnetic powder and a resin containing the magnetic powder. Inductance can be increased by the magnetic



powder. The resin is, for example, epoxy-based, phenol-based, liquid crystal polymer-based, polyimide-based, or acrylic-based, or an organic insulating material made of a mixture thereof. The magnetic powder is, for example, a FeSi-based alloy such as FeSiCr, a FeCo-based alloy, a Fe-based alloy such as NiFe, or an amorphous alloy thereof. Accordingly, in comparison with a magnetic layer made of ferrite, since it is possible to improve DC superposition characteristics by the magnetic powder, and the magnetic powders are insulated from each other by the resin, a loss (iron loss) at a high frequency is reduced. Note that, the magnetic layer may be ferrite or a sintered body of magnetic powder, that contains no organic resin.

(35) The coil **15** extends along the first main surface **10a** between the first magnetic layer **11** and the second magnetic layer **12**. Specifically, the first magnetic layer **11** is present in the reverse Z direction of the coil **15**, and the second magnetic layer **12** is present in the forward Z direction of the coil **15** and in a direction orthogonal to the forward Z direction.

(36) The coil **15** has a coil wiring portion **20**, a first pad portion **21**, and a second pad portion **22**. The coil wiring portion **20** has a first end portion **20a** and a second end portion **20b**. The first pad portion **21** is connected to the first end portion **20a**. The second pad portion **22** is connected to the second end portion **20b**.

(37) The coil wiring portion **20** extends along a circumferential direction with the number of turns equal to or less than 1. As described above, since the number of turns of the coil wiring portion **20** is equal to or less than 1, it is possible to suppress magnetic saturation, to increase a DC superposition allowable current, and to cause a large current to flow. The number of turns of the coil wiring portion **20** is preferably equal to or greater than 0.5. The coil wiring portion **20** is disposed so as to be symmetrical with respect to a center line in the X direction of the element body **10**. The first end portion **20a** is disposed on a side of the first side surface **10c** and a side of the third side surface **10e** of the element body **10**, and the second end portion **20b** is disposed on a side of the second side surface **10d** and a side of the third side surface **10e** of the element body **10**. In this embodiment, the coil wiring portion **20** is formed in a substantially C-shape.

(38) The coil wiring portion **20** has the same width along an extending direction. The width is a dimension in a direction orthogonal to the extending direction. The first end portion **20a** is a portion connected to the first pad portion **21**. The second end portion **20b** is a portion connected to the second pad portion **22**. A line width of the coil wiring portion **20** is less than a line width of the first pad portion **21** and a line width of the second pad portion **22**. Note that, the line width of the first pad portion **21** and the line width of the second pad portion **22** may be the same as the line width of the coil wiring portion **20**.

(39) The first pad portion **21** is a portion to which the first columnar wiring **31** is connected. The first pad portion **21** is a portion overlapping the first columnar wiring **31** when viewed from a direction orthogonal to the first main surface **10a**, has a shape corresponding to an outer peripheral surface of the first columnar wiring **31**, and is slightly greater than the first columnar wiring **31**. The first pad portion **21** extends in a direction that intersects a direction L (hereinafter also referred to as a connection direction L) in which the first end portion **20a** and the second end portion **20b** are connected and that is a direction in which an inner peripheral surface **20c** of the coil wiring portion **20** is approached when viewed from the direction orthogonal to the first main surface **10a**. The connection direction L is a direction in which a center in a width direction of an end edge of the first end portion **20a** and a center in a width direction of an end edge of the second end portion **20b** are connected, when viewed from the direction orthogonal to the first main surface **10a**. The end edge of the first end portion **20a** and the end edge of the second end portion **20b** are each indicated by a dotted line in FIG. 1.

(40) In this embodiment, the first pad portion **21** extends in a direction orthogonal to the connection direction L. The connection direction L is parallel to the X direction, and the direction orthogonal to the connection direction L is parallel to the Y direction. That is, the first pad portion **21** is disposed on a side of the first side surface **10c** of the element body **10**, and extends along the first

side surface **10c** of the element body **10**. The first pad portion **21** is formed in a linear shape when viewed from the direction orthogonal to the first main surface **10a**. Note that, the first pad portion **21** may extend in a direction intersecting the connection direction L at an angle other than a right angle.

(41) The width of the first pad portion **21** is greater than the width of the coil wiring portion **20**. The width of the first pad portion **21** is the same except for an end portion connected to the first end portion **20a** of the coil wiring portion **20**. A width of the end portion of the first pad portion **21** gradually increases along an extending direction of the first pad portion **21**.

(42) The second pad portion **22** is a portion to which the second columnar wiring **32** is connected. The second pad portion **22** is a portion overlapping the second columnar wiring **32** when viewed from the direction orthogonal to the first main surface **10a**, has a shape corresponding to an outer peripheral surface of the second columnar wiring **32**, and is slightly greater than the second columnar wiring **32**. The second pad portion **22** extends in a direction that intersects the direction L (hereinafter also referred to as the connection direction L) in which the first end portion **20a** and the second end portion **20b** are connected and that is a direction in which the inner peripheral surface **20c** of the coil wiring portion **20** is approached when viewed from the direction orthogonal to the first main surface **10a**.

(43) In this embodiment, the second pad portion **22** extends in the direction orthogonal to the connection direction L. That is, the second pad portion **22** is disposed on a side of the second side surface **10d** of the element body **10**, and extends along the second side surface **10d** of the element body **10**. The second pad portion **22** is formed in a linear shape when viewed from the direction orthogonal to the first main surface **10a**. Note that, the second pad portion **22** may extend in a direction intersecting the connection direction L at an angle other than a right angle.

(44) A width of the second pad portion **22** is the same as the width of the first pad portion **21**, and greater than the width of the coil wiring portion **20**. The width of the second pad portion **22** is the same except for an end portion connected to the second end portion **20b** of the coil wiring portion **20**. A width of the end portion of the second pad portion **22** gradually increases along an extending direction of the second pad portion **22**.

(45) The first pad portion **21** is connected to the first columnar wiring **31**, and the second pad portion **22** is connected to the second columnar wiring **32**. The first columnar wiring **31** and the second columnar wiring **32** extend respectively from both ends of the coil **15** in the direction orthogonal to the first main surface **10a**.

(46) Here, in this embodiment, the first pad portion **21** and the second pad portion **22** are not included in a turn of the coil **15**. This is because the first and second pad portions **21** and **22** are not portions that affect a wiring length of the coil **15**. More specifically, a current flowing through the first pad portion **21** flows through the first columnar wiring **31** as is, and a current flowing through the second pad portion **22** flows through the second columnar wiring **32** as is. In other words, a current flowing through the first pad portion **21** flows along the first columnar wiring **31** in the direction orthogonal to the first main surface **10a**, rather than flowing along the circumferential direction of the coil wiring portion **20**, and a current flowing through the second pad portion **22** flows along the second columnar wiring **32** in the direction orthogonal to the first main surface **10a**, rather than flowing along the circumferential direction of the coil wiring portion **20**.

(47) A thickness of each of the coil wiring portion **20**, the first pad portion **21**, and the second pad portion **22** is preferably equal to or greater than 40  $\mu\text{m}$ , and equal to or less than 120  $\mu\text{m}$ , for example. As a working example of the coil wiring portion **20**, the thickness is 35  $\mu\text{m}$ , and a wiring width is 50  $\mu\text{m}$ .

(48) Each of the coil wiring portion **20**, the first pad portion **21**, and the second pad portion **22** is made of a conductive material, for example, made of a metal material having low electric resistance such as Cu, Ag, Au, or Al. In the present embodiment, the inductor component **1** includes only one layer of the coil wiring portion **20**, and thus a reduction in height of the inductor

component **1** can be realized. Note that, the coil wiring portion may have a two-layer configuration of a seed layer and an electrolytic plating layer, and may contain Ti or Ni as the seed layer.

(49) A first extended wiring **201** is connected to a part on a side in the reverse X direction of the coil wiring portion **20**, and the first extended wiring **201** is exposed from the first side surface **10c**. A second extended wiring **202** is connected to a part on a side in the forward X direction of the coil wiring portion **20**, and the second extended wiring **202** is exposed from the second side surface **10d**.

(50) The first extended wiring **201** and the second extended wiring **202** are each a wiring to be connected to a power supply wiring when electrolytic plating is additionally performed after forming a shape of the coil wiring portion **20** in a manufacturing process of the inductor component **1**. In an inductor substrate state before the inductor component **1** is made as an individual piece, by using the power supply wiring, electrolytic plating can be easily performed additionally, and a line width of the coil **15** can be adjusted. In addition, by providing the first extended wiring **201** and the second extended wiring **202**, it is possible to secure strength in cutting of the element body **10** when the inductor component **1** is made as an individual piece, and to improve yield at the time of manufacturing.

(51) Each of the first columnar wiring **31** and the second columnar wiring **32** extends from the coil **15** in the Z direction, and penetrates an inside of the second magnetic layer **12**. The columnar wiring is an example of a “vertical wiring” described in the claims.

(52) The first columnar wiring **31** extends from an upper surface of the first pad portion **21** to the first main surface **10a** of the element body **10**, and an end surface of the first columnar wiring **31** is exposed from the first main surface **10a** of the element body **10**. The second columnar wiring **32** extends from an upper surface of the second pad portion **22** to the first main surface **10a** of the element body **10**, and an end surface of the second columnar wiring **32** is exposed from the first main surface **10a** of the element body **10**.

(53) Thus, the first columnar wiring **31** and the second columnar wiring **32** linearly extend from the coil **15** to the first main surface **10a** in the direction orthogonal to the first main surface **10a**. Thus, each of the first external terminal **41** and the second external terminal **42** can be connected to the coil **15** with a shorter distance, and the inductor component **1** can be reduced in resistance, and increased in inductance. The first columnar wiring **31** and the second columnar wiring **32** are made of a conductive material, and for example, made of the same material as that of the coil **15**.

(54) When viewed from the direction orthogonal to the first main surface **10a**, the first columnar wiring **31** has substantially the same size as that of the first pad portion **21**, and has a shape corresponding to the first pad portion **21**. The second columnar wiring **32** has substantially the same size as that of the second pad portion **22**, and has a shape corresponding to the second pad portion **22**.

(55) Note that, when an upper surface of the coil **15** is covered with the insulating layer **60**, each of the first columnar wiring **31** and the second columnar wiring **32** may be electrically connected to the coil **15** through via wiring that penetrates the insulating layer **60**. The via wiring is a conductor having a line width (diameter, sectional area) less than that of the columnar wiring. In this case, a “vertical wiring” described in the claims is configured with the via wiring and the columnar wiring.

(56) At this time, the first pad portion **21** is a portion to which a first vertical wiring (the first columnar wiring **31** and the via wiring) is connected. The first pad portion **21** includes not only the portion connected to the via wiring but also a portion overlapping the first columnar wiring **31** when viewed from the direction orthogonal to the first main surface **10a**. That is, the first pad portion **21** has a shape corresponding to an outer peripheral surface of the first vertical wiring, when viewed from the direction orthogonal to the first main surface **10a**, and is slightly greater than the first vertical wiring.

(57) The second pad portion **22** is a portion to which a second vertical wiring (the second columnar wiring **32** and the via wiring) is connected. The second pad portion **22** includes not only the portion

connected to the via wiring but also a portion overlapping the second columnar wiring **32** when viewed from the direction orthogonal to the first main surface **10a**. That is, the second pad portion **22** has a shape corresponding to an outer peripheral surface of the second vertical wiring when viewed from the direction orthogonal to the first main surface **10a**, and is slightly greater than the second vertical wiring.

(58) Preferably, a thickness of the coil **15** is less than a thickness of each of the first columnar wiring **31** and the second columnar wiring **32**. According to this, the volume of the element body **10** can be increased, and the inductance can be increased.

(59) Preferably, a thickness of the first magnetic layer **11** is greater than a thickness of the second magnetic layer **12**. According to this, since there is a limit in a thickness related to a process such as a resist thickness for each of the first columnar wiring **31** and the second columnar wiring **32**, there is a limit in the thickness of the second magnetic layer **12**. On the other hand, inductance can be increased by increasing the thickness of the first magnetic layer **11**.

(60) The first external terminal **41** and the second external terminal **42** are provided on the first main surface **10a** of the element body **10**. The first external terminal **41** and the second external terminal **42** are each made of a conductive material, and have a three-layer configuration in which, for example, Cu with low electric resistance and excellent in stress resistance, Ni excellent in corrosion resistance, and Au excellent in wettability and reliability are arranged in this order from an inner side to an outer side.

(61) The first external terminal **41** is in contact with an end surface of the first columnar wiring **31** exposed from the first main surface **10a** of the element body **10**, and is electrically connected to the first columnar wiring **31**. Thus, the first external terminal **41** is electrically connected to a first end (first pad portion **21**) of the coil **15**. The second external terminal **42** is in contact with an end surface of the second columnar wiring **32** exposed from the first main surface **10a** of the element body **10**, and is electrically connected to the second columnar wiring **32**. Thus, the second external terminal **42** is electrically connected to a second end (the second pad portion **22**) of the coil **15**.

(62) The first external terminal **41** is in contact with the first main surface **10a** and the end surface of the first columnar wiring **31**. The second external terminal **42** is in contact with the first main surface **10a** and the end surface of the second columnar wiring **32**. According to this, since the first external terminal **41** spreads across the first main surface **10a** and the first columnar wiring **31**, and the second external terminal **42** spreads across the first main surface **10a** and the second columnar wiring **32**, sizes of the respective first and second external terminals **41** and **42** can be controlled, and design of the first and second external terminals **41** and **42** can be selected from a viewpoint of mounting properties and mechanical strength.

(63) Each of the first external terminal **41** and the second external terminal **42** is preferably in contact with the magnetic powder of the element body **10**. According to this, adhesion strength between each of the first external terminal **41** and the second external terminal **42**, and the element body **10** is improved.

(64) The insulating layer **60** covers a lower surface of the coil **15**. Thus, insulation properties of the coil **15** can be improved. The insulating layer **60** is made of an insulating material that does not contain a magnetic substance. The insulating layer **60** is, for example, an organic resin such as an epoxy resin, a phenol resin, a polyimide resin, a liquid crystal polymer, or a combination thereof, a sintered body such as glass or alumina, or a thin film such as a silicon oxide film, a silicon nitride film, or a silicon oxynitride film.

(65) The coating film **50** is made of an insulating material. The material of the coating film **50** is, for example, the same as the material of the insulating layer **60**. The coating film **50** exposes respective end surfaces of the first and second external terminals **41** and **42**. With the coating film **50**, a short circuit between the first external terminal **41** and the second external terminal **42** can be suppressed. Preferably, the coating film **50** contains a black colorant. With this, it is possible to hide scratches on the surface of the element body **10**, and to improve a rate of good appearance. The

black colorant is, for example, a colorant to which a number including black is assigned in a color index, and specific examples thereof include carbon black-based pigments such as carbon black, ketjen black, perylene black, and titanium oxide, and graphite-based pigments, iron oxide-based pigments, cobalt oxide-based pigments, and anthraquinone-based pigments.

(66) According to the above-described configuration, the first pad portion **21** extends in a direction that intersects the connection direction **L** and that is a direction in which the inner peripheral surface **20c** of the coil wiring portion **20** is approached, and the second pad portion **22** extends in a direction that intersects the connection direction **L** and that is a direction in which the inner peripheral surface **20c** of the coil wiring portion **20** is approached. According to this, it is possible to prevent the first pad portion **21** and the second pad portion **22** from approaching each other, and to reduce a possibility of a short circuit between the first pad portion **21** and the second pad portion **22** having a greatest potential difference in the coil **15**. In addition, it is possible to prevent the first pad portion **21** and the second pad portion **22** from extending outward in the radial direction of the coil **15**, to reduce the inductor component **1** in size. Additionally, an area of the first pad portion **21** can be increased, and connection reliability between the first pad portion **21** and the first columnar wiring **31** can be improved.

(67) On the other hand, for example, when the first pad portion and the second pad portion are extended in the connection direction **L**, the first pad portion and the second pad portion come close to each other and are highly likely to be short-circuited. In addition, when the first pad portion and the second pad portion are extended outward in the radial direction of the coil, it is necessary to increase the element body in size, and the inductor component is increased in size.

(68) FIG. **3** is a plan view of the inductor component. In FIG. **3**, for convenience, only the element body **10** and the coil **15** are drawn, and these are illustrated by solid lines. As illustrated in FIG. **3**, when viewed from the direction orthogonal to the first main surface **10a**, the first pad portion **21** has a linear first plane **211** that is a distal end surface in the extending direction of the first pad portion **21**, and faces the inner peripheral surface **20c** of the coil wiring portion **20**. A shortest distance between the first plane **211** and the inner peripheral surface **20c** of the coil wiring portion **20** is not a minimum among lengths between an outer surface of the first pad portion **21** and the inner peripheral surface **20c** of the coil wiring portion **20**. According to this, it is possible to reduce a possibility of a short circuit between the first plane **211** of the first pad portion **21** and the inner peripheral surface **20c** of the coil wiring portion **20**.

(69) To be specific, when viewed from the direction orthogonal to the first main surface **10a**, the first pad portion **21** has the linear first plane **211** extending in the connection direction **L** and facing the inner peripheral surface **20c** of the coil wiring portion **20**, and the linear second plane **212** extending in the direction orthogonal to the connection direction **L** and facing the inner peripheral surface **20c** of the coil wiring portion **20**. The first plane **211** refers to a surface at a location closest to the inner peripheral surface **20c** of the coil wiring portion **20** among surfaces extending in parallel to the connection direction **L**. The second plane **212** refers to a surface at a location closest to the inner peripheral surface **20c** of the coil wiring portion **20** among surfaces extending in the direction orthogonal to the connection direction **L**.

(70) A surface parallel to the first plane **211** and tangent to a part, of the inner peripheral surface **20c** of the coil wiring portion **20**, facing the first plane **211** is defined as a first virtual surface **S211**, and a surface parallel to the second plane **212** and tangent to a part, of the inner peripheral surface **20c** of the coil wiring portion **20**, facing the second plane **212** is defined as a second virtual surface **S212**. At this time, a first distance **a** between the second virtual surface **S212** and the second plane **212** is less than a second distance **b** between the first virtual surface **S211** and the first plane **211**.

(71) According to the above-described configuration, since the first distance **a** is less than the second distance **b**, it is possible to reduce a possibility of a short circuit between the first plane **211** of the first pad portion **21** and a part, of the inner peripheral surface **20c** of the coil wiring portion **20**, facing the first plane **211**. Even if a short circuit occurs between the second virtual surface **S212**

and the second plane **212**, a variation in a coil length is suppressed as compared to a case where a short circuit occurs between the first virtual surface **S211** and the first plane **211**, and thus influence of a layer short can be suppressed.

(72) Note that, the first distance *a* may be equal to or greater than the second distance *b*. In this case, the first pad portion **21** can be increased in the extending direction thereof. Thus, a size of the first pad portion **21** can be increased, and the connection reliability between the first pad portion **21** and the first columnar wiring **31** can be improved.

(73) Also, the second pad portion **22** may have a similar configuration to that of the first pad portion **21**, and has a similar effect to that of the first pad portion **21**. That is, when viewed from the direction orthogonal to the first main surface **10a**, the second pad portion **22** has a linear first plane **221** that is a distal end surface in the extending direction of the second pad portion **22** and that faces the inner peripheral surface **20c** of the coil wiring portion **20**. A shortest distance between the first plane **221** and the inner peripheral surface **20c** of the coil wiring portion **20** is not a minimum among lengths between an outer surface of the second pad portion **22** and the inner peripheral surface **20c** of the coil wiring portion **20**.

(74) To be specific, the second pad portion **22** has the linear first plane **221** extending in the connection direction *L* and facing the inner peripheral surface **20c** of the coil wiring portion **20**, and a linear second plane **222** extending in the direction orthogonal to the connection direction *L* and facing the inner peripheral surface **20c** of the coil wiring portion **20**. A surface parallel to the first plane **221** and tangent to a part, of the inner peripheral surface **20c** of the coil wiring portion **20**, facing the first plane **221** is defined as a first virtual surface **S221**, and a surface parallel to the second plane **222** and tangent to a part, of the inner peripheral surface **20c** of the coil wiring portion **20**, facing the second plane **222** is defined as a second virtual surface **S222**. At this time, the first distance *a* between the second virtual surface **S222** and the second plane **222** is less than the second distance *b* between the first virtual surface **S221** and the first plane **221**. Note that, the first distance *a* may be equal to or greater than the second distance *b*.

(75) As illustrated in FIG. 3, in the first pad portion **21**, it is preferable that the first distance *a* between the second virtual surface **S212** and the second plane **212** be equal to or greater than twice the particle size *D50* of the magnetic powder.

(76) Here, unless otherwise specified, the particle size *D50* of the magnetic powder is measured from a scanning electron microscope (SEM) image of a cross-section of a central portion in a longitudinal direction of the element body **10** of the inductor component **1**. At this time, the SEM image preferably includes 10 or more of the magnetic powders, and is acquired at a magnification of 2000 times, for example. The SEM image as described above is acquired at three or more locations from the above-described cross-section, the magnetic powder and the others are classified by binarization or the like, an equivalent circle diameter of the magnetic powder in the SEM image is calculated, and an intermediate value (median diameter) when the equivalent circle diameters are arranged in order of magnitude is defined as the particle size *D50* of the magnetic powder.

(77) According to the above-described configuration, by increasing the distance between the second virtual surface **S212** and the second plane **212** to be greater than the magnetic powder, it is possible to further reduce a possibility of a short circuit between the second virtual surface **S212** and the second plane **212**.

(78) Also, the second pad portion **22** may have a similar configuration to that of the first pad portion **21**, and has a similar effect to that of the first pad portion **21**. That is, in the second pad portion **22**, the first distance *a* between the second virtual surface **S222** and the second plane **222** is equal to or greater than twice the particle size *D50* of the magnetic powder.

(79) As illustrated in FIG. 3, preferably, in the first pad portion **21**, the shortest distance *c* between the first pad portion **21** and the second pad portion **22** is greater than the first distance *a* between the second virtual surface **S212** and the second plane **212** when viewed from the direction orthogonal to the first main surface **10a**.

(80) According to the above-described configuration, it is possible to further reduce the possibility of a short circuit between the first pad portion **21** and the second pad portion **22** having the greatest potential difference in the coil **15**. Even if a short circuit occurs between the second virtual surface **S212** and the second plane **212**, a variation in the coil length is suppressed compared to a case where a short circuit occurs between the first pad portion **21** and the second pad portion **22**, and thus influence of a layer short can be suppressed.

(81) Note that, the shortest distance  $c$  may be equal to or less than the first distance  $a$ . In this case, since the first pad portion **21** and the second pad portion **22** can be brought close to each other, a length of the coil wiring portion **20** can be increased. Thus, the coil length can be increased, and the inductance can be increased.

(82) Also, the second pad portion **22** may have a similar configuration to that of the first pad portion **21**, and has a similar effect to that of the first pad portion **21**. That is, in the second pad portion **22**, the shortest distance  $c$  is greater than the first distance  $a$ . Note that, the shortest distance  $c$  may be equal to or less than the first distance  $a$ .

(83) As illustrated in FIG. 3, the shortest distance  $c$  between the first pad portion **21** and the second pad portion **22** is preferably equal to or greater than twice the particle size  $D_{50}$  of the magnetic powder. According to the above-described configuration, by increasing the distance between the first pad portion **21** and the second pad portion **22** to be greater than the magnetic powder, it is possible to further reduce the possibility of a short circuit between the first pad portion **21** and the second pad portion **22**.

(84) As illustrated in FIG. 3, preferably, in the first pad portion **21**, the first distance  $a$  is less than the second distance  $b$ , and the second distance  $b$  is less than the shortest distance  $c$ . For example, the first distance  $a$  is 200  $\mu\text{m}$ , the second distance  $b$  is 210  $\mu\text{m}$ , and the shortest distance  $c$  is 720  $\mu\text{m}$ . According to the above-described configuration, since the shortest distance  $c$  is greater than the first distance  $a$  and the second distance  $b$ , it is possible to further reduce the possibility of a short circuit between the first pad portion **21** and the second pad portion **22** having the greatest potential difference. Note that, the second distance  $b$  may be less than the shortest distance  $c$  regardless of a magnitude relation between the first distance  $a$  and the second distance  $b$ .

(85) Also, the second pad portion **22** may have a similar configuration to that of the first pad portion **21**, and has a similar effect to that of the first pad portion **21**.

#### Manufacturing Method

(86) Next, a manufacturing method of the inductor component **1** will be described. FIG. 4A to FIG. 4M correspond to the A-A section (FIG. 2) in FIG. 1.

(87) As illustrated in FIG. 4A, a base substrate **70** is prepared. The base substrate **70** is made of an inorganic material such as ceramic, glass, or silicon, for example. An insulating layer **71** is applied to the base substrate **70**. As illustrated in FIG. 4B, a predetermined pattern is formed and cured on the insulating layer **71** using a photolithography method, to form the insulating layer **60**.

(88) As illustrated in FIG. 4C, a seed layer (Ti/Cu) (not illustrated) is formed on a main surface of the base substrate **70** including an upper surface of the insulating layer **60** by a known method such as a sputtering method or a vapor deposition method. Thereafter, a DFR (dry film resist) **75** is attached, and an opening having a predetermined shape is formed in the DFR **75** by using a photolithography method. Then, by using an electrolytic plating method, while feeding electricity to the seed layer, the coil wiring portion **20**, the first pad portion **21**, and the second pad portion **22** are formed on the insulating layer **60**.

(89) As illustrated in FIG. 4D, the DFR **75** is peeled off. At this time, the seed layer may be used for electrolytic plating of a columnar wiring without etching the seed layer, or the seed layer may be etched to form a seed layer for forming a columnar wiring again.

(90) As illustrated in FIG. 4E, the DFR **75** is attached again, and an opening having a predetermined shape is formed in the DFR **75** by using a photolithography method. The openings are through-holes respectively corresponding to a location where the first columnar wiring **31** is

provided on the first pad portion **21**, and a location where the second columnar wiring **32** is provided on the second pad portion **22**. Using electrolytic plating, the first columnar wiring **31** is provided on the first pad portion **21**, and the second columnar wiring **32** is provided on the second pad portion **22**. As illustrated in FIG. 4F, the DFR **75** is peeled off and the seed layer is etched.

(91) As illustrated in 4G, a magnetic sheet to be the second magnetic layer **12** is pressure-bonded from above the main surface of the base substrate **70** toward the coil wiring portion **20**, and the coil wiring portion **20**, the first pad portion **21**, the second pad portion **22**, the first columnar wiring **31**, and the second columnar wiring **32** are covered with the second magnetic layer **12**. Thereafter, an upper surface of the second magnetic layer **12** is ground, and the end surface of the first columnar wiring **31** and the end surface of the second columnar wiring **32** are exposed from the upper surface of the second magnetic layer **12**.

(92) As illustrated in FIG. 4H, the coating film **50** is applied to the upper surface of the second magnetic layer **12**. Then, the coating film **50** is formed into a predetermined pattern and cured by using a photolithography method. The predetermined pattern has a shape having an opening at a location corresponding to the first external terminal **41** and an opening at a location corresponding to the second external terminal **42**.

(93) As illustrated in FIG. 4I, the base substrate **70** is removed by polishing. At this time, a part or all of the insulating layer **60** may be removed. As illustrated in FIG. 4J, another magnetic sheet to be the first magnetic layer **11** is pressure-bonded from below the coil wiring portion **20** toward the coil wiring portion **20**, and the coil wiring portion **20** is covered with the first magnetic layer **11**. Thereafter, the first magnetic layer **11** is ground to have a predetermined thickness.

(94) As illustrated in FIG. 4K, the first external terminal **41** and the second external terminal **42** are formed in the openings of the coating film **50**, respectively. The first external terminal **41** is in contact with the end surface of the first columnar wiring **31** and the upper surface of the second magnetic layer **12**. The second external terminal **42** is in contact with the end surface of the second columnar wiring **32** and the upper surface of the second magnetic layer **12**. Further, the coating film **50** is applied to a lower surface of the first magnetic layer **11**, and cured.

(95) As illustrated in FIG. 4L, an inductor component is made as an individual piece along cut lines D, and as illustrated in FIG. 4M, the inductor component **1** is manufactured.

#### WORKING EXAMPLES

(96) Next, working examples will be described.

(97) FIG. 5A illustrates an inductor component **100A** of a first comparative example, and FIG. 5B illustrates an inductor component **100B** of a second comparative example. FIG. 5C illustrates an inductor component **1A** of a first working example, and FIG. 5D illustrates an inductor component **1B** of a second working example. In each of FIG. 5A to FIG. 5D, only an element body, a coil, and external terminals are illustrated for convenience, the element body and the coil are indicated by solid lines, and the external terminals are indicated by two dot chain lines. Although a columnar wiring is omitted, the columnar wiring has substantially the same size as that of a pad portion when viewed from the Z direction, and has a shape corresponding to the pad portion.

(98) As illustrated in 5A, in the inductor component **100A** of the first comparative example, each of a first pad portion **121A** and a second pad portion **122A** is quadrangular, and has a size of 100  $\mu\text{m}$ ×100  $\mu\text{m}$ . A distance between a first external terminal **141A** and a second external terminal **142A** is 200  $\mu\text{m}$ .

(99) As illustrated in FIG. 5B, in the inductor component **100B** of the second comparative example, a first pad portion **121B** extends in the forward X direction as compared to the first pad portion **121A** in FIG. 5A, and a second pad portion **122B** extends in the reverse X direction as compared to the second pad portion **122A** in FIG. 5A. Hereinafter, a length difference in the X direction between the first pad portion **121B** and the first pad portion **121A**, and a length difference in the X direction between the second pad portion **122B** and the second pad portion **122A** are each referred to as an extension amount of a pad portion. A distance between a first external terminal **141B** and a



second external terminal **142B** is the same as the distance between the first external terminal **141A** and the second external terminal **142A** in FIG. 5A.

(100) As illustrated in FIG. 5C, in the inductor component **1A** of the first working example, a first pad portion **21A** extends in the forward Y direction as compared to the first pad portion **121A** in FIG. 5A, and a second pad portion **22A** extends in the forward Y direction as compared to the second pad portion **122A** in FIG. 5A. Hereinafter, a length difference in the Y direction between the first pad portion **21A** and the first pad portion **121A**, and a length difference in the Y direction between the second pad portion **22A** and the second pad portion **122A** are each referred to as an extension amount of a pad portion. A distance between a first external terminal **41A** and a second external terminal **42A** is the same as the distance between the first external terminal **141A** and the second external terminal **142A** in FIG. 5A.

(101) As illustrated in FIG. 5D, in the inductor component **1B** of the second working example, a first pad portion **21B** has the same shape as that of the first pad portion **21A** in FIG. 5C, and a second pad portion **22B** has the same shape as that of the second pad portion **22A** in FIG. 5C. A distance between a first external terminal **41B** and a second external terminal **42B** is 600  $\mu\text{m}$ , which is greater than the distance between the first external terminal **41A** and the second external terminal **42A** in FIG. 5C.

(102) FIG. 6 shows a relationship between relative values of an L value and extension amounts of pad portions in the above first comparative example, the above second comparative example, the above first working example, and the above second working example. FIG. 7 is a graph of an enlarged part of FIG. 6.

(103) In FIG. 6 and FIG. 7, an x mark denotes the first comparative example, a triangle mark denotes the second comparative example, a square mark denotes the first working example, and a circle mark denotes the second working example. A relative value of an L value indicates a relative value when an L value of the first comparative example is defined as 1. As described above, an extension amount of a pad portion refers to a length difference from a pad portion of the first comparative example.

(104) As shown in FIG. 6 and FIG. 7, in each of the second comparative example and the first working example, the relative value of the L value slightly decreased as the extension amount of the pad portion increased. It is considered that this is because volume of the magnetic layer decreases. The decrease in the relative value of the L value is within an allowable range at the time of manufacturing, and there is no problem in quality. On the other hand, in each of the second comparative example and the first working example, as the extension amount of the pad portion increases, an area of the columnar wiring also increases, and as a result, a contact area with the external terminal increases, and connection reliability is improved.

(105) In the second working example, the relative value of the L value is greater, as compared to the first working example. This is because, since the distance between the two external terminals in the second working example is greater than the distance between the two external terminals in the first working example, it is possible to reduce obstruction of a flow of magnetic flux of the coil by the external terminals in the second working example.

(106) As described above, in each of the first comparative example, the second comparative example, the first working example, and the second working example, the L value can be ensured. However, in the first comparative example, since the pad portion is small, a contact area with the external terminal is small, and connection reliability is lowered. Further, in the second comparative example, there is a high possibility that the two pad portions come close to each other and are short-circuited. On the other hand, in the first working example and the second working example, since a contact area with the external terminal is increased, connection reliability is improved, and since the two pad portions do not come close to each other, a possibility of a short circuit can be reduced. Furthermore, in the second working example, it is possible to reduce obstruction of a flow of magnetic flux and to suppress a decrease in the L value.

## Second Embodiment

(107) FIG. 8 is a plan view illustrating a second embodiment of the inductor component. In FIG. 8, for convenience, only an element body and a coil are drawn, and these are illustrated by solid lines. The second embodiment is different from the first embodiment in a configuration of a pad portion of a coil. This different configuration will be described below. The other configurations are the same as those of the first embodiment, and are denoted by the same reference numerals as those of the first embodiment, and description thereof will be omitted.

(108) As illustrated in FIG. 8, in an inductor component 1C of the second embodiment, a first pad portion 21C of a coil 15C has a first portion 21a, a second portion 21b, and a corner portion 21c between the first portion 21a and the second portion 21b when viewed from a direction orthogonal to the first main surface 10a. The corner portion 21c is an extension of the first portion 21a and is included in the first portion 21a.

(109) The first portion 21a extends from the first end portion 20a in the connection direction L. The second portion 21b extends from a side of the first portion 21a in a direction that intersects the connection direction L and that is a direction in which the inner peripheral surface 20c of the coil wiring portion 20 is approached. Each of the first portion 21a and the second portion 21b is a portion extending with the same width. In FIG. 8, for convenience, a boundary between each of the first portion 21a and the second portion 21b, and the corner portion 21c is indicated by a two dot chain line. In this embodiment, the second portion 21b extends in a direction orthogonal to the connection direction L. That is, the first pad portion 21C is formed in an L-shape when viewed from the direction orthogonal to the first main surface 10a.

(110) A first columnar wiring connected to the first pad portion 21C has a shape corresponding to the first pad portion 21C when viewed from the direction orthogonal to the first main surface 10a. Thus, an area of the first pad portion 21C can be increased, and connection reliability between the first pad portion 21C and the first columnar wiring can be improved.

(111) When viewed from the direction orthogonal to the first main surface 10a, the first portion 21a of the first pad portion 21C protrudes inward in a radial direction of the coil 15C from the inner peripheral surface 20c of the coil wiring portion 20. The first plane 211 is a surface at a location closest to the inner peripheral surface 20c of the coil wiring portion 20 among surfaces extending in parallel to the connection direction L of the first pad portion 21C, and thus corresponds to a surface on a distal end side of the second portion 21b. The second plane 212 is a surface at a location closest to the inner peripheral surface 20c of the coil wiring portion 20 among surfaces extending in the direction orthogonal to the connection direction L of the first pad portion 21C, and thus corresponds to a surface on a base end side (side of the first end portion 20a) of the first portion 21a. In this case, the first distance a between the second virtual plane S212 and the second plane 212 can be easily made less than the second distance b between the first virtual surface S211 and the first plane 211. The first distance a may be the same as the second distance b, and for example, the first distance a is 100  $\mu\text{m}$ , the second distance b is 100  $\mu\text{m}$ , and the shortest distance c is 200  $\mu\text{m}$ .

(112) Similarly, a second pad portion 22C of the coil 15C has a first portion 22a, a second portion 22b, and a corner portion 22c between the first portion 22a and the second portion 22b when viewed from the direction orthogonal to the first main surface 10a.

(113) The first portion 22a extends from the second end portion 20b in the connection direction L. The second portion 22b extends from a side of the first portion 22a in a direction that intersects the connection direction L and that is a direction in which the inner peripheral surface 20c of the coil wiring portion 20 is approached. In this embodiment, the second portion 21b extends in the direction orthogonal to the connection direction L. In other words, the second pad portion 22C is formed in an L-shape when viewed from the direction orthogonal to the first main surface 10a, and is formed symmetrically with the first pad portion 21C.

(114) A second columnar wiring connected to the second pad portion 22C has a shape

corresponding to the second pad portion **22C** when viewed from the direction orthogonal to the first main surface **10a**. Thus, an area of the second pad portion **22C** can be increased, and connection reliability between the second pad portion **22C** and the second columnar wiring can be improved.

(115) When viewed from the direction orthogonal to the first main surface **10a**, the first portion **22a** of the second pad portion **22C** protrudes inward in the radial direction of the coil **15C** from the inner peripheral surface **20c** of the coil wiring portion **20**. The first plane **221** corresponds to a surface on a distal end side of the second portion **22b**. The second plane **222** corresponds to a surface on a base end side (side of the second end portion **20b**) of the first portion **22a**.

#### Third Embodiment

(116) FIG. **9** is a plan view illustrating a third embodiment of the inductor component. In FIG. **9**, for convenience, only an element body and a coil are drawn, and these are illustrated by solid lines. The third embodiment is different from the second embodiment in a configuration of a pad portion of a coil. This different configuration will be described below. The other configurations are the same as those of the second embodiment, and are denoted by the same reference numerals as those of the second embodiment, and description thereof will be omitted. Note that, although the configuration of the pad portion is different from that of the second embodiment, a first portion and a second portion of the pad portion are defined in the same manner as those of the second embodiment.

(117) As illustrated in FIG. **9**, in an inductor component **1D** of the third embodiment, the first portion **21a** of a first pad portion **21D** of a coil **15D** does not protrude inward in a radial direction of the coil **15D** from the inner peripheral surface **20c** of the coil wiring portion **20** when viewed from a direction orthogonal to the first main surface **10a**. In other words, the first portion **21a** protrudes outward in the radial direction from an outer peripheral surface of the coil wiring portion **20**. At this time, the first plane **211** corresponds to a surface on a distal end side of the second portion **21b**, and the second plane **212** corresponds to a side surface of the second portion **21b**.

(118) According to the above-described configuration, it is possible to increase a distance between the inner peripheral surface **20c** of the coil wiring portion **20** and the first pad portion **21D** in the connection direction **L**, and to further reduce a possibility of a short circuit between the coil wiring portion **20** and the first pad portion **21D** in the connection direction **L**. For example, the first distance **a** between the second virtual plane **S212** and the second plane **212** can be easily made greater than the second distance **b** between the first virtual surface **S211** and the first plane **211**. For example, the first distance **a** is 200  $\mu\text{m}$ , the second distance **b** is 100  $\mu\text{m}$ , and the shortest distance **c** is 200  $\mu\text{m}$ .

(119) Similarly, the first portion **22a** of a second pad portion **22D** of the coil **15D** does not protrude inward in the radial direction of the coil **15D** from the inner peripheral surface **20c** of the coil wiring portion **20** when viewed from the direction orthogonal to the first main surface **10a**. In other words, the first portion **22a** protrudes outward in the radial direction from the outer peripheral surface of the coil wiring portion **20**. At this time, the first plane **221** corresponds to a surface on a distal end side of the second portion **22b**, and the second plane **222** corresponds to a side surface of the second portion **22b**.

(120) According to the above-described configuration, it is possible to increase a distance between the inner peripheral surface **20c** of the coil wiring portion **20** and the second pad portion **22D** in the connection direction **L**, and to further reduce a possibility of a short circuit between the coil wiring portion **20** and the second pad portion **22D** in the connection direction **L**.

#### Fourth Embodiment

(121) FIG. **10** is a sectional view illustrating an embodiment of a mounting component. As illustrated in FIG. **10**, a mounting component **3** includes a substrate **300** and an inductor component **1E** disposed in the substrate **300**.

(122) The inductor component **1E** has the configuration of the inductor component **1** illustrated in

the first embodiment and does not include the first external terminal **41**, the second external terminal **42**, and the coating film **50**.

(123) The substrate **300** has a main body portion **310**, an internal wiring **320**, and an external wiring **330**. The main body portion **310** is made of, for example, an insulating material. The internal wiring **320** is provided in the main body portion **310**. The external wiring **330** is provided on a main surface **310a** of the main body portion **310**. The internal wiring **320** is connected to the first columnar wiring **31** and the second columnar wiring **32** in the inductor component **1E**. The internal wiring **320** is connected to the external wiring **330**. Thus, the inductor component **1E** is electrically connected to the external wiring **330**.

(124) According to the above-described configuration, since the inductor component **1E** is built into the substrate **300**, it is possible to increase a mounting area of a surface of the substrate **300**.

(125) Note that, the present disclosure is not limited to the above-described embodiments, and design changes can be made without departing from the scope of the present disclosure. For example, the features of the first to fourth embodiments may be combined in various ways.

(126) In the above-described embodiments, at least a part of the first pad portion extends in the direction orthogonal to the connection direction **L**, but may extend in a direction intersecting the connection direction **L** at an angle other than a right angle. Similarly, at least a part of the second pad portion extends in the direction orthogonal to the connection direction **L**, but may extend in a direction intersecting the connection direction **L** at an angle other than a right angle.

(127) In the above-described embodiments, the shape of each of the first pad portion and the second pad portion is formed in the linear shape or the L-shape, but may be any shape as long as the shape has a part extending in a direction, that intersects the connection direction **L** and that is a direction in which the inner peripheral surface of the coil wiring portion is approached.

## Claims

1. An inductor component, comprising: an element body having a main surface; a coil disposed in the element body, and extending along the main surface; and a first vertical wiring and a second vertical wiring in the element body such that respective end surfaces thereof are exposed from the main surface of the element body, and connected to respective ends of the coil, the coil including a coil wiring portion having a first end portion and a second end portion, and extending along a circumferential direction, with a number of turns equal to or less than 1, a first pad portion to which the first vertical wiring is connected and which is connected to the first end portion, and a second pad portion to which the second vertical wiring is connected and which is connected to the second end portion, wherein when viewed from a direction orthogonal to the main surface, at least a portion of the first pad portion extends in a direction that intersects a direction in which the first end portion and the second end portion of the coil wiring portion are connected and extends in a direction close towards an inner peripheral surface of the coil wiring portion, and when viewed from the direction orthogonal to the main surface, at least a portion of the second pad portion extends in the direction that intersects the direction in which the first end portion and the second end portion of the coil wiring portion are connected and extends in the direction close towards the inner peripheral surface of the coil wiring portion.

2. The inductor component according to claim 1, further comprising: a first external terminal in contact with the main surface of the element body and the end surface of the first vertical wiring; and a second external terminal in contact with the main surface of the element body and the end surface of the second vertical wiring.

3. The inductor component according to claim 2, wherein the element body includes a magnetic powder, and the first external terminal and the second external terminal are in contact with the magnetic powder.

4. The inductor component according to claim 1, further comprising: a coating film on the main

surface of the element body, wherein the coating film contains a black colorant.

5. The inductor component according to claim 1, wherein when viewed from the direction orthogonal to the main surface of the element body, the first pad portion has a linear first plane that is a distal end surface in an extending direction of the first pad portion and that faces the inner peripheral surface of the coil wiring portion, and a shortest distance between the first plane and the inner peripheral surface of the coil wiring portion is not a minimum among distances between an outer surface of the first pad portion and the inner peripheral surface of the coil wiring portion.

6. The inductor component according to claim 1, wherein when viewed from the direction orthogonal to the main surface, the first pad portion has a linear first plane extending in the direction in which the first end portion and the second end portion of the coil wiring portion are connected, and facing the inner peripheral surface of the coil wiring portion, and a linear second plane extending in a direction orthogonal to the direction in which the first end portion and the second end portion of the coil wiring portion are connected, and facing the inner peripheral surface of the coil wiring portion, and when a first virtual surface is defined as a surface parallel to the first plane of the first pad portion and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing the first plane, and a second virtual surface is defined as a surface parallel to the second plane of the first pad portion and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing the second plane, and a first distance between the second virtual surface and the second plane is less than a second distance between the first virtual surface and the first plane.

7. The inductor component according to claim 1, wherein when viewed from the direction orthogonal to the main surface, the first pad portion has a linear first plane extending in the direction in which the first end portion and the second end portion of the coil wiring portion are connected, and facing the inner peripheral surface of the coil wiring portion, and a linear second plane extending in a direction orthogonal to the direction in which the first end portion and the second end portion of the coil wiring portion are connected, and facing the inner peripheral surface of the coil wiring portion, and when a first virtual surface is defined as a surface parallel to the first plane of the first pad portion and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing the first plane, and a second virtual surface is defined as a surface parallel to the second plane of the first pad portion and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing the second plane, a first distance between the second virtual surface and the second plane is equal to or greater than a second distance between the first virtual surface and the first plane.

8. The inductor component according to claim 6, wherein the element body includes a magnetic powder, and the first distance between the second virtual surface and the second plane is equal to or greater than twice a particle size D50 of the magnetic powder.

9. The inductor component according to claim 6, wherein when viewed from the direction orthogonal to the main surface, a shortest distance between the first pad portion and the second pad portion is greater than the first distance between the second virtual surface and the second plane.

10. The inductor component according to claim 6, wherein when viewed from the direction orthogonal to the main surface, a shortest distance between the first pad portion and the second pad portion is equal to or less than the first distance between the second virtual surface and the second plane.

11. The inductor component according to claim 9, wherein the element body includes a magnetic powder, and the shortest distance between the first pad portion and the second pad portion is equal to or greater than twice a particle size D50 of the magnetic powder.

12. The inductor component according to claim 1, wherein when viewed from the direction orthogonal to the main surface, the first pad portion has a first portion extending from the first end portion of the coil wiring portion in the direction in which the first end portion and the second end portion of the coil wiring portion are connected, and a second portion extending from the first portion side in the direction that intersects the direction in which the first end portion and the

second end portion of the coil wiring portion are connected and extending in the direction close towards the inner peripheral surface of the coil wiring portion, and when viewed from the direction orthogonal to the main surface, the second pad portion has a first portion extending from the second end portion of the coil wiring portion in the direction in which the first end portion and the second end portion of the coil wiring portion are connected, and a second portion extending from the first portion side in the direction that intersects the direction in which the first end portion and the second end portion of the coil wiring portion are connected and extending in the direction close towards the inner peripheral surface of the coil wiring portion.

13. The inductor component according to claim 12, wherein when viewed from the direction orthogonal to the main surface, the first portion of the first pad portion does not protrude inward from the inner peripheral surface of the coil wiring portion.

14. The inductor component according to claim 1, wherein a thickness of the coil is less than a thickness of each of the first vertical wiring and the second vertical wiring.

15. The inductor component according to claim 1, wherein the element body has a first magnetic layer and a second magnetic layer stacked in order in a first direction orthogonal to the main surface, the first vertical wiring and the second vertical wiring extend in the first direction from the respective ends of the coil, the first magnetic layer is present in a direction opposite to the first direction of the coil, the second magnetic layer is present in the first direction of the coil and in a direction orthogonal to the first direction, the first vertical wiring and the second vertical wiring penetrate the second magnetic layer, and a thickness of the first magnetic layer is greater than a thickness of the second magnetic layer.

16. A mounting component, comprising: a substrate; and the inductor component according to claim 1 disposed in the substrate.

17. The inductor component according to claim 2, further comprising: a coating film on the main surface of the element body, wherein the coating film contains a black colorant.

18. The inductor component according to claim 2, wherein when viewed from the direction orthogonal to the main surface of the element body, the first pad portion has a linear first plane that is a distal end surface in an extending direction of the first pad portion and that faces the inner peripheral surface of the coil wiring portion, and a shortest distance between the first plane and the inner peripheral surface of the coil wiring portion is not a minimum among distances between an outer surface of the first pad portion and the inner peripheral surface of the coil wiring portion.

19. The inductor component according to claim 2, wherein when viewed from the direction orthogonal to the main surface, the first pad portion has a linear first plane extending in the direction in which the first end portion and the second end portion of the coil wiring portion are connected, and facing the inner peripheral surface of the coil wiring portion, and a linear second plane extending in a direction orthogonal to the direction in which the first end portion and the second end portion of the coil wiring portion are connected, and facing the inner peripheral surface of the coil wiring portion, and when a first virtual surface is defined as a surface parallel to the first plane of the first pad portion and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing the first plane, and a second virtual surface is defined as a surface parallel to the second plane of the first pad portion and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing the second plane, and a first distance between the second virtual surface and the second plane is less than a second distance between the first virtual surface and the first plane.

20. The inductor component according to claim 2, wherein when viewed from the direction orthogonal to the main surface, the first pad portion has a linear first plane extending in the direction in which the first end portion and the second end portion of the coil wiring portion are connected, and facing the inner peripheral surface of the coil wiring portion, and a linear second plane extending in a direction orthogonal to the direction in which the first end portion and the second end portion of the coil wiring portion are connected, and facing the inner peripheral surface of the coil wiring portion, and when a first virtual surface is defined as a surface parallel to the first

plane of the first pad portion and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing the first plane, and a second virtual surface is defined as a surface parallel to the second plane of the first pad portion and tangent to a part, of the inner peripheral surface of the coil wiring portion, facing the second plane, a first distance between the second virtual surface and the second plane is equal to or greater than a second distance between the first virtual surface and the first plane.

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